Pràctica 3
Dashboard de monitorització CPD

Costin Sebastian Oprea Parvu

Iker Miranda Arjona

Dani Soldevila

SMX 2 B

Projecte Intermodular

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Objectiu de la pràctica

Aquesta pràctica consisteix a convertir una placa ESP32 en un servidor web capaç de monitoritzar una sala de servidors (CPD). El sistema llegeix dades de sensors (temperatura i llum), les mostra en una pàgina web i activa alertes visuals i un LED quan es detecten valors perillosos. També inclou una pàgina secundària amb crèdits i l'ús de GitHub per gestionar el desenvolupament.

Esquema de connexions (fritzing)

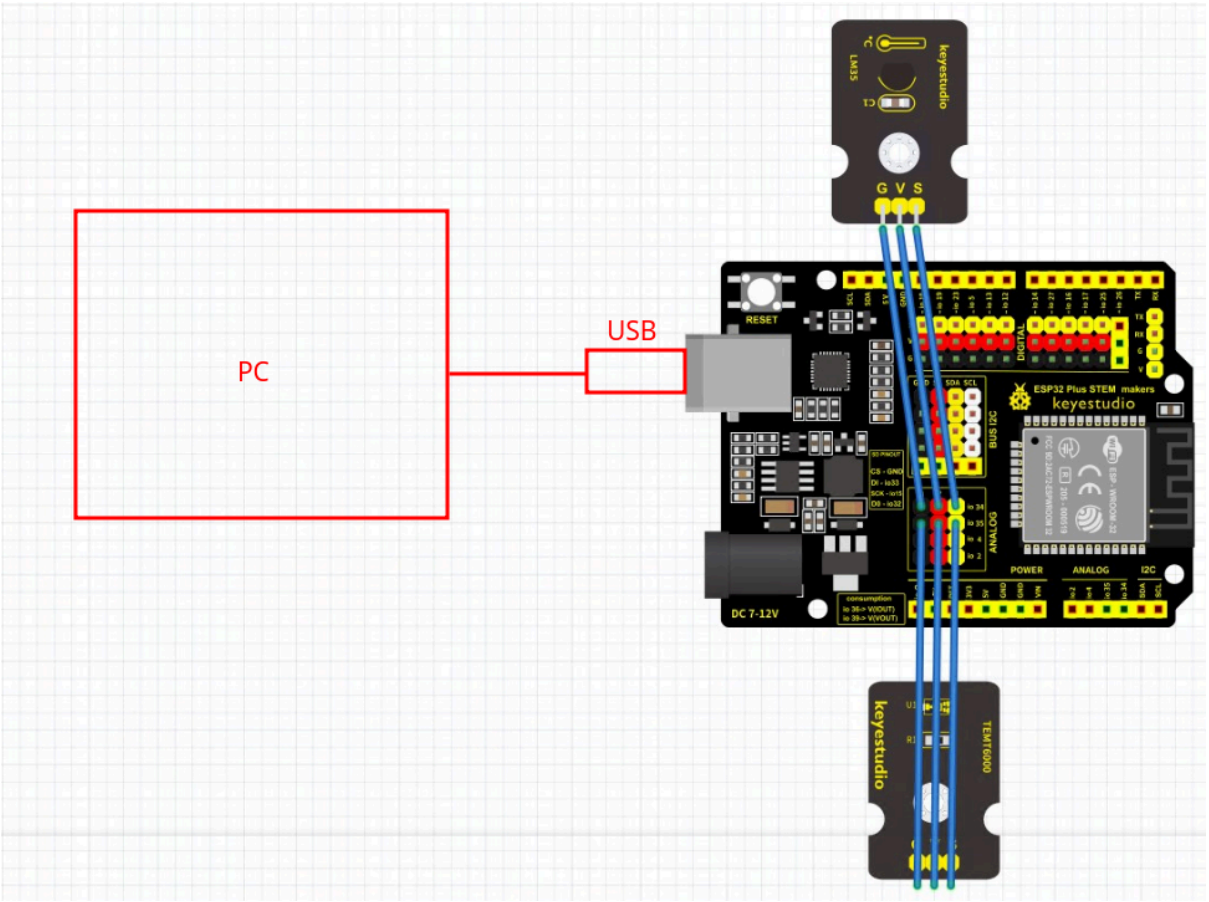
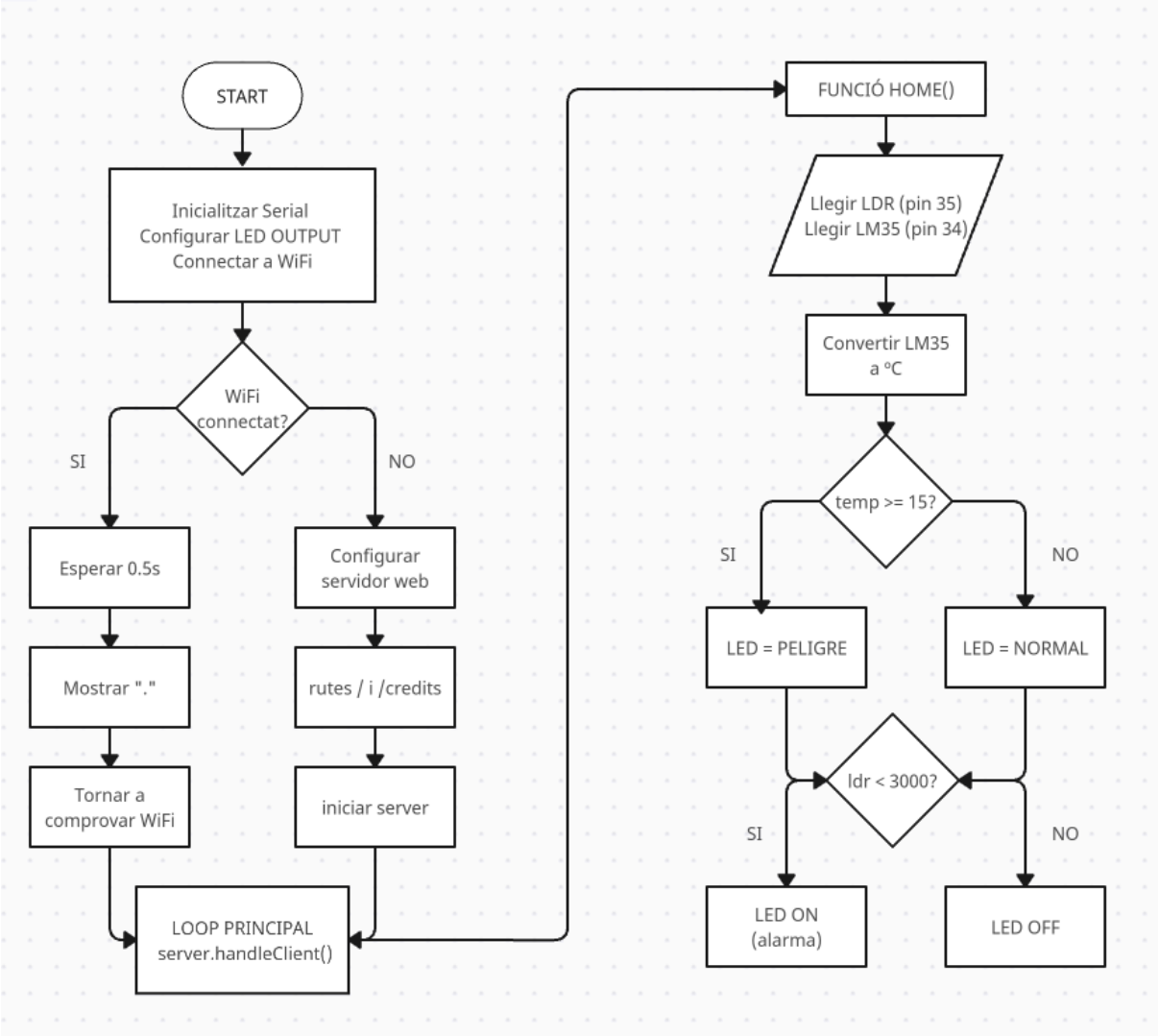
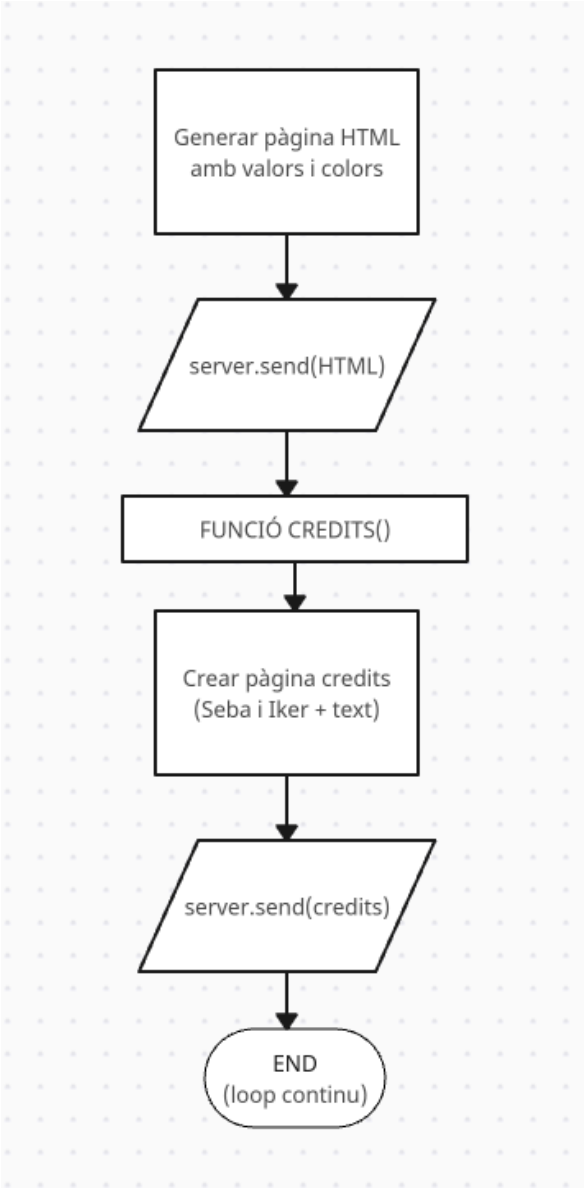


Diagrama de flux del programa (com fer un diagrama de flux)



https://app.creately.com/d/3yvxbNq7BG0/edit



https://app.creately.com/d/3yvxbNq7BG0/edit

Les 2 imatges són el mateix diagrama. El que passa és que la pàgina era premium i no et deixava posar més coses

Proves

Issue 1:

```
#include <WiFi.h>

const char* ssid = "ProjectesSMX";
const char* password = "Hola1234";

void setup() {
  Serial.begin(115200);

  WiFi.begin(ssid, password);
  Serial.print("Connectant...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnectat!");
  Serial.print("IP: ");
  Serial.println(WiFi.localIP());
}

void loop() {
}
```

/dev/ttyUSB0

```
ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4
Connectant.......
Connectat!
IP: 192.168.137.127
```

Issue 2:

sketch_mar12a

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "ProjectesSMX";
const char* password = "Hola1234";

WebServer server(80);

void enviarPaginaPrincipal() {
  server.send(200, "text/plain", "Hola, soc el teu ESP32");
}

void setup() {
  Serial.begin(115200);

  WiFi.begin(ssid, password);
  Serial.print("Connectant...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnectat!");
  Serial.print("IP: ");
  Serial.println(WiFi.localIP());

  server.on("/", enviarPaginaPrincipal);
  server.begin();
}

void loop() {
  server.handleClient();
}
```

/dev/ttyUSB0

```
ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4
Connectant......
Connectat!
IP: 192.168.137.70
```

Issue 3:

sketch_mar12a

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "ProjectesSMX";
const char* password = "Hola1234";

WebServer server(80);

void pagina() {
  String html = "<html><head><meta charset='UTF-8'>";
  html += "<style>";
  html += "body{font-family:sans-serif;text-align:center;}";
  html += ".card{padding:20px;background:gray;color:white;display:inline-block;margin:10px;border-radius:10px;}";
  html += "</style></head><body>";

  html += "<h1>Monitor CPD</h1>";

  html += "<div class='card'>";
  html += "<h3>LDR</h3>";
  html += "<p>---</p>";
  html += "</div>";

  html += "<div class='card'>";
  html += "<h3>LM35</h3>";
  html += "<p>--- °C</p>";
  html += "</div>";

  html += "</body></html>";

  server.send(200, "text/html", html);
}

void setup() {
  Serial.begin(115200);

  WiFi.begin(ssid, password);
  Serial.print("Connectant...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnectat!");
  Serial.println(WiFi.localIP());

  server.on("/", pagina);
  server.begin();
}

void loop() {
  server.handleClient();
}
```

/dev/ttyUSB0

```
ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4
Connectant.......
Connectat!
IP: 192.168.137.70
```

Issue 4:

sketch_mar12a

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "ProjectesSMX";
const char* password = "Hola1234";

const int pinLDR = 35;
const int pinLM35 = 34;

WebServer server(80);

void pagina() {
  int ldr = analogRead(pinLDR);
  float tempC = (analogRead(pinLM35) * 3.3 / 4095.0) * 100.0;

  String html = "<html><head><meta charset='UTF-8'>";
  html += "<meta http-equiv='refresh' content='5'>";
  html += "<style>";
  html += "body{font-family:sans-serif;text-align:center;}";
  html += ".card{padding:20px;background:gray;color:white;display:inline-block;margin:10px;border-radius:10px;}";
  html += "</style></head><body>";

  html += "<h1>Monitor CPD</h1>";

  // LDR
  html += "<div class='card'>";
  html += "<h3>LDR</h3>";
  html += "<p>" + String(ldr) + "</p>";
  html += "</div>";

  // LM35
  html += "<div class='card'>";
  html += "<h3>LM35</h3>";
  html += "<p>" + String(tempC, 1) + " °C</p>";
  html += "</div>";

  html += "</body></html>";

  server.send(200, "text/html", html);
}

void setup() {
  Serial.begin(115200);

  WiFi.begin(ssid, password);
  Serial.print("Connectant...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnectat!");
  Serial.println(WiFi.localIP());

  server.on("/", pagina);
  server.begin();
}
```

```
/dev/ttyUSB0

ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4
Connectant.......
Connectat!
IP: 192.168.137.70
```

Issue 5 i 6:

sketch_mar12a

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "ProjectesSMX";
const char* password = "Hola1234";

const int pinLDR = 35;
const int pinLM35 = 34;
const int LED = 27;

WebServer server(80);

void pagina() {
  int ldr = analogRead(pinLDR);
  float tempC = (analogRead(pinLM35) * 3.3 / 4095.0) * 100.0;

  String colorTemp = "#28a745";
  String estadoTemp = "NORMAL";

  if (tempC >= 15.0) {
    colorTemp = "#dc3545";
    estadoTemp = "PELIGRO";
  }

  String colorLDR = "#28a745";
  String estadoLDR = "NORMAL";

  if (ldr < 3000) {
    colorLDR = "#dc3545";
    estadoLDR = "PELIGRO";
  }

  if (tempC >= 15.0 || ldr < 3000) {
    digitalWrite(LED, HIGH);
  } else {
    digitalWrite(LED, LOW);
  }
```

```
  String html = "<html><head><meta charset='UTF-8'>";
  html += "<meta http-equiv='refresh' content='5'>";
  html += "<style>body{font-family:sans-serif;text-align:center;}.card{padding:20px;color:white;display:inline-block;margin:10px;border-radius:10px;}</style>"
  html += "</head><body>";

  html += "<h1>Monitor CPD</h1>";

  html += "<div class='card' style='background:" + colorTemp + ";'>";
  html += "<h3>Temperatura</h3>";
  html += "<p>" + String(tempC, 1) + " °C</p>";
  html += "<p>" + estadoTemp + "</p>";
  html += "</div>";

  html += "<div class='card' style='background:" + colorLDR + ";'>";
  html += "<h3>LDR</h3>";
  html += "<p>" + String(ldr) + "</p>";
  html += "<p>" + estadoLDR + "</p>";
  html += "</div>";

  html += "</body></html>";

  server.send(200, "text/html", html);
}

void setup() {
  Serial.begin(115200);

  pinMode(LED, OUTPUT);

  WiFi.begin(ssid, password);
  Serial.print("Connectant...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnectat!");
  Serial.println(WiFi.localIP());

  server.on("/", pagina);
  server.begin();
}
```

```
void loop() {
  server.handleClient();
}
```

/dev/ttyUSB0

```
ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4
Connectant.......
Connectat!
IP: 192.168.137.70
```

Issue 7:

sketch_apr23a §

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "ProjectesSMX";
const char* password = "Hola1234";

const int pinLDR = 35;
const int pinLM35 = 34;
const int LED = 27;

WebServer server(80);

void home() {
  int ldr = analogRead(pinLDR);
  float tempC = (analogRead(pinLM35) * 3.3 / 4095.0) * 100.0;

  String colorTemp = "#28a745";
  String estadoTemp = "NORMAL";

  if (tempC >= 15.0) {
    colorTemp = "#dc3545";
    estadoTemp = "PELIGRO";
  }

  String colorLDR = "#28a745";
  String estadoLDR = "NORMAL";

  if (ldr < 3000) {
    colorLDR = "#dc3545";
    estadoLDR = "PELIGRO";
  }

  if (tempC >= 15.0 || ldr < 3000) digitalWrite(LED, HIGH);
  else digitalWrite(LED, LOW);

  String html = "<html><head><meta charset='UTF-8'>";
  html += "<meta http-equiv='refresh' content='5'>";
  html += "<style>";
  html += "body{font-family:sans-serif;text-align:center;}";
  html += ".card{padding:20px;color:white;display:inline-block;margin:10px;border-radius:10px;}";
  html += "a{display:block;margin-top:20px;font-size:18px;}";
  html += "</style></head><body>";

  html += "<h1>Monitor CPD</h1>";

  html += "<div class='card' style='background:" + colorTemp + ";'>";
  html += "<h3>Temperatura</h3>";
  html += "<p>" + String(tempC, 1) + " °C</p>";
  html += "<p>" + estadoTemp + "</p>";
  html += "</div>";

  html += "<div class='card' style='background:" + colorLDR + ";'>";
  html += "<h3>LDR</h3>";
  html += "<p>" + String(ldr) + "</p>";
  html += "<p>" + estadoLDR + "</p>";
  html += "</div>";

  html += "<a href='/credits'>Veure Credits</a>";

  html += "</body></html>";

  server.send(200, "text/html", html);
}

void credits() {
  String html = "<html><head><meta charset='UTF-8'>";
  html += "<style>body{font-family:sans-serif;text-align:center;}</style>";
  html += "</head><body>";

  html += "<h1>Seba i Iker</h1>";
  html += "<p>Aquest projecte utilitza una ESP32 per monitoritzar temperatura i llum en temps real.</p>";
  html += "<p>Quan els valors superen els límits, es genera una alerta i s'activa un LED.</p>";

  html += "<a href='/'>Tornar</a>";

  html += "</body></html>";

  server.send(200, "text/html", html);
}
```

```
void setup() {
  Serial.begin(115200);

  pinMode(LED, OUTPUT);

  WiFi.begin(ssid, password);
  Serial.print("Connectant...");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnectat!");
  Serial.print("IP: ");
  Serial.println(WiFi.localIP());

  server.on("/", home);
  server.on("/credits", credits);

  server.begin();
}

void loop() {
  server.handleClient();
}
```

```
ets Jun  8 2016 00:22:57

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4876
ho 0 tail 12 room 4
load:0x40078000,len:16560
load:0x40080400,len:3500
entry 0x400805b4
Connectant......
Connectat!
IP: 172.16.61.158
```

Enllaç al Vídeo demostratiu del funcionament

https://docs.google.com/videos/d/1BZdE_8Vq2Kb3aVkQmWhA3DIPLtbUmsoO7IV-FynrQrk/edit?scene=id.p#scene=id.p